TETHER: Tethered Causal Temporal Refinement for Frozen Surgical Stereo

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Abstract— **Modern stereo estimators are applied independently to each frame and can fluctuate under specularities, weak texture, tissue deformation, tool occlusions and camera motion. Video stereo methods fix this by building temporal reasoning into the estimator, so a new backbone means a new temporal model. We ask the opposite question: can one causal temporal module be trained once and attached, unchanged, to frozen and heterogeneous stereo estimators?**

We present TETHER, a causal plug-and-play refiner that answers it. One frozen $154{,}874$-parameter checkpoint, trained on three estimators and one dataset, improves *every* estimator we evaluate on *every* arena we evaluate: five architectures — two of them excluded from its training — across development, held-out and out-of-domain splits, fifteen backbone–arena cells without an exception. Zero-shot on DRENDS, a dataset no part of the module has seen, disparity EPE, Bad3 and metric depth improve in all 25 recording–backbone cells, by -5.40, -19.51, -4.02 and -3.47%, with nothing adapted, retrained or tuned for the domain. The two unseen estimators are not distinguishable from the three seen ones, which is the claim.

The module reads only what a black box emits: preceding disparity motion-aligned with frozen SEA-RAFT, and a shared 38-channel evidence space of disparity, motion and image geometry — no learned appearance features, no frozen backbone, no future frame — feeding a reset-and-fusion head that predicts a bounded interpolation between the current estimate and that aligned memory. Under a matched state machine the gain is not temporal smoothing: 3.69% mean EPE against 0.59% for the best fixed or EMA blend, which buys its mean error with bad pixels while the learned gate does not. We additionally introduce a refinement-centric protocol scoring introduced error alongside recovered error under fixed, prediction-independent support, and apply it to our own claims.

I. Introduction

Metric three-dimensional perception underpins camera navigation, tool–tissue distance estimation and deformation analysis in robot-assisted surgery, and stereo endoscopy is attractive because depth comes from images the platform already provides. It is also hard: endoscopic scenes carry weak texture, non-Lambertian tissue, specularities, blood, smoke, deformable anatomy, rapid viewpoint change and tool occlusion [18], [19]. Modern estimators [11]–[15] give increasingly strong frame-wise disparity, but nothing in a frame-wise model requires its predictions to cohere over time. Video stereo methods build temporal reasoning into the model [1]–[4], at the price of a temporal architecture, model-specific representations, future observations, or co-design with the estimator.

We study a different deployment setting: *can one causal temporal module be trained once and attached unchanged to heterogeneous frozen stereo estimators?* Backbone and temporal logic can then evolve independently, a new estimator replacing the previous one without the module needing its cost volume, hidden state, confidence head or training pipeline. We keep one inductive bias from the video-stereo literature — decide whether motion-aligned history should influence the current estimate, and constrain the correction to a convex interpolation between the two — and redesign everything around it for a black-box interface.

Does the learned head add anything simple temporal averaging would not? Matched fixed fusion, EMA, warped-memory propagation and forward–backward confidence weighting recover only a small fraction of TETHER's gain, while a ground-truth oracle shows the aligned memory carries far more than the policy exploits. The core problem is therefore not obtaining temporal evidence but deciding how much of it to use — and Sec. VI shows the head decides *how much* well and *where* no better than chance, which is what leaves the oracle gap open. A second question is how such an intervention should be evaluated: smoothness is not correctness, and lower mean EPE hides whether refinement introduces new bad pixels or rare catastrophic frames, so we evaluate relative to the frozen prediction and separate geometry, temporal fidelity, benefit, harm and tail behaviour.

Our contributions are three.

(i) **One module, every backbone, every domain.** We measure a single frozen checkpoint across the full backbone $\times$ domain matrix — five estimators, two excluded from its training, over development, held-out and out-of-domain splits, including the joint unseen-backbone, unseen-domain cell usually left unmeasured — and it improves all fifteen cells, with the unseen estimators indistinguishable from the seen ones.

(ii) **A black-box interface that makes that possible.** We formulate surgical temporal stereo refinement as a *causal plug-and-play adaptation problem* over frozen, heterogeneous estimators, needing neither future frames nor backbone internals, and instantiate it as TETHER: a 154,874-parameter reset-and-fusion head over causal SEA-RAFT alignment, a shared 38-channel external evidence space, support-confined convex fusion and bounded corrected-state recurrence. Under a matched state machine the gain is not temporal smoothing.

(iii) **A protocol that scores the intervention, not the prediction.** We measure introduced error alongside recovered error under fixed, prediction-independent support, and it changes conclusions: every spatially uniform blend recovering a meaningful share of the gain buys it by *raising* the bad-pixel rate, and the one unlearned rule that does not is the one that varies in space — which is the paper's thesis arriving

from the other direction. Applied to our own claims the protocol is equally unkind: degenerate history-replay wins the no-reference metric, and the head's own weight inverts its relation to harm across splits.

II. Related Work

a) Deep stereo matching.: Stereo has moved from cost-volume regression to iterative recurrent estimation [11], [12] and broad-domain zero-shot models [13]–[15]. Our work is orthogonal: we neither modify the estimator nor assume an architecture.

b) Temporal stereo and depth.: CODD introduced the reset/fusion mechanism we build on [1]; DynamicStereo [2], PPMStereo [4] and TC-Stereo [6] follow, all co-designed with the estimator, so a new backbone means a new temporal model. Match Stereo Videos exposes BiDAStabilizer as a plug-in on a frozen estimator [3], but propagates bidirectionally. A parallel line enforces consistency on monocular video depth by diffusion [24] or stabiliser networks [16]: a different output space, none causal at clip level. A further line adapts the stereo weights online [25], the opposite of our constraint.

c) What can actually be compared.: We are aware of no published method that is simultaneously causal, black-box and estimator-agnostic. CODD [1], whose two-gate decision we build on, is causal but not black-box: its fusion network consumes matching features from its own stereo network, which a cached third-party backbone does not expose. The 142-channel configuration of Sec. V-J began as the closest reproduction of it our interface admits — its projection replaced by our warp, and a frozen ResNet-18 standing in for the features it would have taken from the estimator — so the comparison against it in Table III is the nearest thing to running CODD here. Of the black-box candidates, StereoDiffusion [7] ships no implementation, and neural disparity refinement [8] and blind video temporal consistency [9] publish code whose weights are no longer retrievable; the supplement audits the set. Two methods are runnable here and we report both: BiDAStabilizer [3], which refines a frozen estimator as we do but bidirectionally, and TC-Stereo [6], an integrated architecture propagating its state by a rigid reprojection from per-frame pose — a privileged input our interface does not assume but SCARED-C supplies, so we run it as its authors intended (Sec. V-I).

d) Evaluating temporal refinement.: Stereo benchmarks report frame-wise geometry and video methods add consistency measures, but neither characterises an intervention: a prediction can grow smoother without growing more correct, and a lower mean error can hide newly created failures. Recovered and introduced error must both be measured, which connects to selective prediction [17] and to the uncertainty literature [26]. Our contribution is not a new metric but a common protocol under fixed, prediction-independent support.

e) Surgical stereo geometry.: SCARED [18] and DRENDS [22] supply the endoscopic stereo and time-of-flight geometry we evaluate on; D4D [21] enters only as the substrate for a no-reference control; SERV-CT [19] not at all, its pairs being non-consecutive, so temporal refinement on it is an exact identity. Downstream reconstruction and tracking pipelines consume exactly the per-frame geometry we refine [20], [27], which is why we treat refinement as an intervention on that geometry.

III. Method

A. External Causal Interface

A frozen estimator S produces positive-left disparity d_t^{raw} and its validity M_t^{raw}. The temporal module never reads the internal representation of S; it consumes only externally observable quantities — RGB, disparity, validity, optical flow and evidence derived from them. At time t the cache holds the preceding left image, the preceding raw disparity and validity, the previous fused state and its age; nothing from $t+1$ onwards exists. The map to learn is $d_t^{\mathrm{fused}} = T_\theta(d_t^{\mathrm{raw}}, m_{t-1}, I_t^L, I_t^R, I_{t-1}^L)$.

B. Causal Motion Alignment

Frozen SEA-RAFT [5], [23] estimates both directions between the two already observed left images; the reverse direction is used only to measure reliability, and bidirectional *flow estimation* between two observed frames is not bidirectional temporal inference. Previous geometry is pulled onto the current grid, $\widetilde{m}_t(x) = m_{t-1}(x + F_{t\to t-1}(x))$, by bilinear sampling with zero padding; warp support $\mathcal{S}_t$ records whether the source coordinate lies inside the preceding image, and historical validity is sampled on the same grid. Pulling the reverse flow the same way gives the forward–backward residual e_t^{FB} and the reliability cue $C_t^{FB} = \mathcal{S}_t \exp(-e_t^{FB}/\rho_t)$ with ρ_t scaled by the flow magnitudes. The zero padding is not a detail: it makes $\widetilde{m}_t$ exactly zero outside the source image, and Sec. V-K shows what happens when that value reaches the output.

C. Shared External Evidence

All estimators are mapped into one 38-channel tensor on the cache grid, built from raw disparity, aligned memory, the two images, the previous left image, the flow, C_t^{FB}, the warp support and the sampled historical validity. Twenty-five channels are disparity-geometry correlations formed at quarter resolution — 16 self-correlations of the two candidates and 9 raw-to-aligned cross-correlations over a dilated 3×3 neighbourhood — and 13 are formed on the grid directly: the two disparities, their signed and absolute residual, 6 motion components and 3 RGB. Normalisation is fixed before training. Nothing in this space is learned, no network evaluates it, and no ground truth, backbone identity or future frame enters it. An earlier configuration added 104 quarter-resolution channels from a frozen ResNet-18 [10] for 142 in total; Sec. V-J reports it as an ablation, because it costs $157{,}504$ frozen parameters and three forward passes per frame and does not survive a three-seed comparison. Exact channel construction and constants are in the supplement.

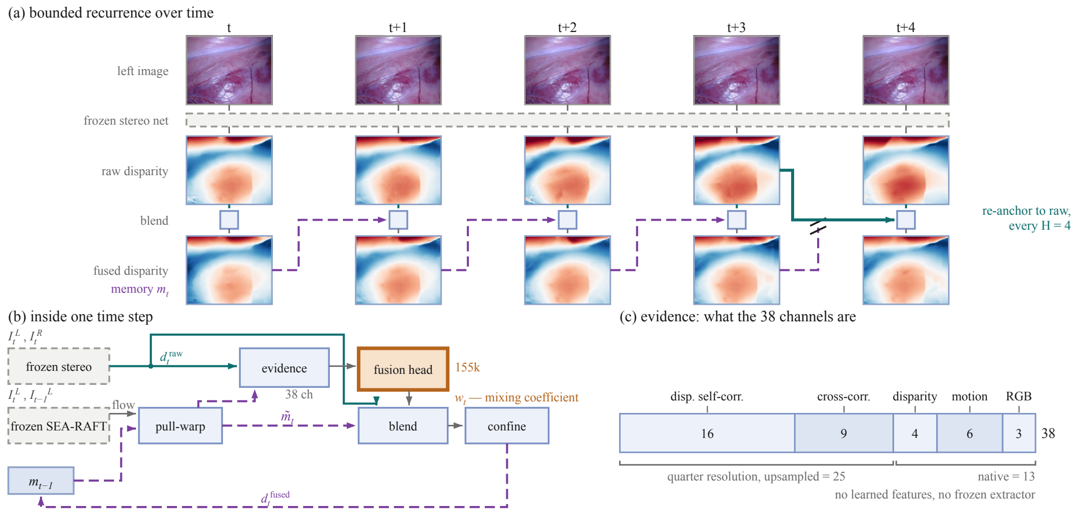

Fig. 1. TETHER at three scales. **(a)** Bounded recurrence: memory carries the *fused* disparity forward, and every $H{=}4$ frames the chain is cut and re-tied to raw geometry the frozen network produced independently (teal). The five frames are consecutive; the two disparity rows share one colour scale and differ by 0.126 px on average, with one fitted plane removed from all ten panels so the anatomy is visible under a 7 px depth ramp. **(b)** One time step; dashed grey is frozen, the single orange block is the only trained component and emits one scalar per pixel, w_t, rather than geometry. **(c)** The evidence space: none of the 38 channels carries ground truth, backbone identity or a future frame.

D. Reset-and-Fusion Decision Head

The two-gate decomposition of the temporal decision, and the constraint that the correction be convex in the two candidates, are CODD's [1]; the architecture, the motion pathway, the evidence space, the supervision and the training are not.

A cache-grid pathway and a quarter-resolution context pathway feed a full-resolution reset branch predicting $r_t = \sigma(z_t^r)$ and a coarse fusion branch predicting $f_t = \uparrow_4 [\sigma(z_t^f)]$, giving the effective temporal weight $w_t = r_t f_t$ from 154,874 trainable parameters. Running the context branch coarsely is a compute choice costing $16\times$ less than full resolution, which Sec. V-J reports as an ablation. Layer widths are in the supplement.

Temporal evidence is only defined where the aligned memory is usable, so the per-pixel *support* is the conjunction of current raw validity, aligned temporal validity and warp support,

$$S_t = M_t^{\text{raw}} \wedge \widetilde{M}_t \wedge \mathcal{S}_t, \tag{1}$$

— the warp support $\mathcal{S}_t$ of the previous paragraph, the raw validity and the sampled historical validity — and confine the intervention to it. Final disparity is

$$\boxed{d_t^{\text{fused}} = \begin{cases} (1-w_t)d_t^{\text{raw}} + w_t\widetilde{m}_t, & S_t = 1, \\ d_t^{\text{raw}}, & S_t = 0. \end{cases}} \tag{2}$$

Because $w_t \in [0,1]$ the fused disparity lies between the two candidates on the support, and off it the module is the exact identity on raw geometry. Applying (1) to the output and not only computing it is load-bearing (Sec. V-K). Reset output bias is initialized to -2 and fusion to zero, so initial intervention is conservative.

E. Bounded Corrected-State Recurrence

After refinement $m_t = d_t^{\text{fused}}$, so alignment and fusion errors can become future evidence. We bound corrected-state lifetime: the first output of a sequence is raw, a re-anchor restores previous raw disparity as state, and otherwise the previous fused output is propagated.

The shipped policy permits four corrected-state uses before a raw re-anchor, so H denotes state lifetime rather than a temporal input window.

The method remains one-step causal: $\{I_{t-1}, I_t, m_{t-1}\} \rightarrow d_t^{\text{fused}}$.

F. Training Objective

For target disparity g_t let $\Delta_t = |\widetilde{m}_t - g_t| - |d_t^{\text{raw}} - g_t|$, so $\Delta_t < 0$ marks pixels where aligned memory is the better candidate; training support is the intersection of GT coverage, current validity, aligned temporal validity and warp support. The unweighted objective $\mathcal{L} = \mathcal{L}_{disp} + \mathcal{L}_{reset} + \mathcal{L}_{fusion}$ pairs a Huber term on the fused disparity with two decision terms supervising each branch against the sign of Δ_t at native margins 5 and 1 px, scaled to the cache grid, plus a weak pull of the fusion weight towards 0.5 where the two candidates are within the fusion margin. No explicit temporal-consistency objective is used: temporal behaviour is a consequence of the decision, never a training target.

IV. Experimental Protocol

A. Training

The temporal refiner is trained on temporally curated SCARED-C data derived from SCARED [18].

Training uses frozen predictions from S^{2}M^2-S [14], RAFT-Stereo [11] and Stereo Anywhere [13]; CREStereo [12] and Fast-FoundationStereo [15] are excluded from temporal-head training, and no backbone identity is ever given to TETHER.

Training uses D1, D3 and D6 with D2 for development and D7 held out, over 8,616 frames giving 8,606 causal pairs per backbone from three seen backbones, in non-overlapping $H = 4$ clips, batch size 4, 12 epochs of AdamW at 2×10^{-4} with weight decay 10^{-4} and gradient clipping 1.0, selecting on development fused EPE. The budget is not a truncation: across three independently seeded runs the selected minimum falls at epoch 8, 11 and 11, every one of them inside the schedule, so the model has converged before the schedule ends and repeating it three times says the same thing. A probe confirms that nothing waits outside it. Retrained under the locked recipe with the epoch budget alone raised and not eligible for selection, run to 47 epochs, it oscillates from epoch 13 onward with a standard deviation of $0.0029\,\mathrm{px}$ inside a band of 0.013 and shows no trend; its own best epoch sits $0.005\,\mathrm{px}$ from that plateau, which is the width of the band, and no later epoch improves on it. Four times the budget changes only noise. One ablation (A4) diverges after epoch 10, which best-validation selection catches.

B. External Evaluation

DRENDS [22] supplies dynamic robotic-endoscopy stereo with calibration and metric geometry; we report five curated recordings, 7476 frames, with all five estimators. Its disparity reference comes from metric ToF geometry through stereo calibration, so it is useful metric evidence but not independent stereo ground truth, and the ToF reference is itself temporally filtered. The zero-shot claim applies to the SCARED-trained checkpoint: no DRENDS data enters training, selection or calibration.

C. Unified Temporal-Refinement Evaluation

Evaluation support cannot depend on the prediction: $M_{\text{eval}} = M_{\text{GT}} \cap M_{\text{protocol}}$, and invalid predictions on valid support are penalised rather than silently removed. The framework scores five dimensions.

a) Spatial geometry.: EPE/MAE, median, RMSE, P90/P95/P99, maximum, invalid rate and Bad-1/3/5 on disparity; Bad-2/5/10 mm, AbsRel, SqRel and $\delta_{1,2,3}$ on metric depth.

b) Temporal fidelity.: At lags $k \in \{1, 2, 4, 8\}$, grid-based temporal change error $\mathrm{DTCE} = |\Delta\text{prediction} - \Delta\text{target}|$ on fixed support, and, given correspondence, motion-compensated $\mathrm{TEPE}_{\text{corr}}$, estimated-flow correspondences marked as proxies. DTCE scores agreement with the *reference* change, not stillness: a frozen prediction has $\Delta{=}0$ and is charged the whole of the reference motion, so the degenerate replay that wins the no-reference score of Sec. VII does not win this one. Every DTCE figure here is at $k{=}4$ unless stated. We do not build on the motion-compensated family: it improves on all five D7 backbones at all four lags, but on D2 its two members disagree beyond one frame.

c) Intervention harm and benefit.: For $\delta e = e^{\text{fused}} - e^{\text{raw}}$ we measure the harm and benefit masses $H^{+} = \mathbb{E}[\max(\delta e, 0)]$, $B^{+} = \mathbb{E}[\max(-\delta e, 0)]$ and their rates HUR, BUR. Per bad-pixel threshold we report NewBad, RecoveredBad, bad-rate delta and identity preservation — the fraction of pixels already within τ still within it afterwards, a retention rate and not a count of untouched pixels, since the blend moves almost every pixel on the support. Per frame we retain the mean signed error change and its tails, and where a risk score exists, risk–coverage analysis [17]. Aggregation is named at every number: macro is the mean of per-cell relative reductions, pooled sums before the ratio.

V. Results

The claim is one module across backbones and domains, so the results are ordered that way: the transfer matrix, the seed evidence that it is stable, why it works, and what it costs.

A. Cross-Backbone Development Behaviour

$H = 4$ improves all five estimators in mean D2 EPE (Table I), from 1.41% to 8.19% averaging per-sequence reductions — 1.70% to 8.04% from the pooled cells the table prints — and all six unseen-backbone sequence groups improve; on D7 the mean excluding Stereo Anywhere is 5.15%. One D2 cell worsens, $\mathrm{S^2M^2}$-S on keyframe 2, by 0.28%.

B. Held-Out Cross-Backbone Confirmation

We next evaluate the already frozen $H = 4$ policy on held-out D7. All five backbones improve, including both unseen during temporal-head training: mean EPE decreases $0.4921 \to 0.4651$, or 5.49% pooling before the ratio and 5.40% averaging per-cell reductions.

The improvement is not restricted to the mean: Bad1, RMSE and P99 decrease for every backbone on both splits, so the EPE gain is not bought by a worse bad-pixel rate or a broader tail. New bad pixels do appear and are measured below. InvalidRate stays zero in all ten SCARED-C backbone–split evaluations, a property of SCARED-C and not of the module (Sec. V-K). The joint unseen-backbone, unseen-domain cell is the lower-right block of Table I.

C. External Zero-Shot Geometry on DRENDS

We evaluate the frozen checkpoint on five DRENDS recordings with all five estimators, two excluded from temporal-head training, 7476 frames each, nothing adapted or tuned for the domain. Disparity EPE, Bad3, metric depth MAE and RMSE improve in *every* one of the 25 recording–backbone cells, by -5.40, -19.51, -4.02 and -3.47% on average; the two unseen estimators are not distinguishable from the three seen ones, which is the claim. Residual variance is set more by the recording than the estimator but not uniformly: the EPE change spans 0.87 points across backbones on Vid10 against 3.29 on Vid13.

Averaged over backbones the mean EPE reduction is 3.96% seen against 2.67% unseen on D2, 5.80 and 4.79% on D7, 5.57 and 5.16% on DRENDS: every combination positive. We read no ordering into it — on D2 the seen advantage is carried entirely by Stereo Anywhere and the

TABLE I

ONE FROZEN TETHER CHECKPOINT, SEED 0, ACROSS FIVE ESTIMATORS AND THREE ARENAS; EVERY EPE AND BAD1 CELL IS RAW→TETHER, LOWER IS BETTER. TWO ESTIMATORS WERE EXCLUDED FROM TEMPORAL-HEAD TRAINING, SO THE LOWER-RIGHT BLOCK IS THE JOINT SHIFT USUALLY LEFT UNMEASURED. DTCE IS THE MACRO REDUCTION IN TEMPORAL CHANGE ERROR AT LAG k=4, HIGHER IS BETTER; ON DEVELOPMENT IT IS BACKBONE-DEPENDENT (SEC. V-F).

		D2 development		D7 held out			DRENDS, out of domain		
Backbone	Status	EPE	Bad1 (%)	EPE	Bad1 (%)	DTCE	EPE	Bad1 (%)	DTCE
S^2M^2-S	seen	.2704 → **.2658**	1.39 → **1.27**	.5176 → **.4992**	6.44 → **6.03**	6.5	1.3477 → **1.2812**	62.55 → **58.98**	11.3
RAFT-Stereo	seen	.2691 → **.2623**	1.60 → **1.43**	.4671 → **.4296**	6.09 → **5.75**	7.3	1.3125 → **1.2371**	58.83 → **55.19**	8.5
Stereo Anywhere	seen	.3022 → **.2779**	2.18 → **1.74**	.4774 → **.4408**	6.23 → **5.77**	7.9	1.3685 → **1.2834**	62.73 → **57.99**	13.9
CREStereo	unseen	.2668 → **.2583**	1.57 → **1.40**	.5165 → **.4912**	6.35 → **6.05**	6.4	1.3027 → **1.2296**	55.17 → **52.21**	11.8
Fast-FoundationStereo	unseen	.2725 → **.2658**	1.60 → **1.44**	.4819 → **.4645**	5.92 → **5.65**	4.3	1.2838 → **1.2228**	54.88 → **52.06**	9.6

ordering reverses without it — and withdraw the claim we drew about which axis costs more. This establishes zero-shot geometric transfer under a ToF-derived reference; it does not establish universal robustness, and DRENDS remains a non-independent reference.

D. Seed Variance, and What It Reveals About Selection

We pre-registered a seed study before any result existed: only the seed varies, resampling is at sequence granularity rather than pixel, and selecting the best seed is prohibited.

Pooled over five backbones the relative Bad1 reduction is $12.73 \pm 0.32\%$ on D2 and $5.84 \pm 0.22\%$ on held-out D7; EPE gives $4.13 \pm 0.38\%$ and $5.61 \pm 0.42\%$, where $\pm$ is the standard deviation over the three seeds throughout this paper and *range* always means min–max. A paired bootstrap over the 15 and 20 backbone–sequence cells gives $[8.96, 16.60]$ and $[4.00, 8.71]$ for Bad1 and $[2.55, 5.67]$ and $[4.03, 7.28]$ for EPE, positive in every resample. We report those intervals as descriptive only: the cells share three and four underlying sequences, so they are not independent draws and the interval is narrower than the evidence warrants. The distribution-free statement the splits can carry is that the effect is positive in $3/3$ and $4/4$ sequences, a sign test at $p = 0.125$ and $p = 0.0625$.

The spread is the strongest argument for the shipped configuration, and not what we found first. The 142-channel ablation is stable held out (5.15 ± 0.34 Bad1) and unstable on development, its seeds falling monotonically over a 5.03-point range. We read that as selection optimism, since checkpoints are selected on D2. Removing the learned evidence removes the instability instead — 0.32 points of D2 seed deviation against 2.53, with no change to the selection rule — so the instability was the evidence space and not the protocol.

E. Does the Learned Temporal Policy Matter?

We isolate the learned decision policy on D2, seed 0. All variants share the frozen predictions, the alignment, the support contract and the evaluation code, and all run unconfined so the comparison shares one implementation (confinement moves D2 EPE by 1.2×10^{-5} px). Figures are pooled over the fifteen backbone–sequence cells, whose raw EPE is 0.276202.

The fifteen policies (Table II, full grid in the supplement) are the horizon sweep and continuous recurrence for the learned head, fixed $w \in \{0.1, 0.2, 0.3, 0.5\}$ rising monotonically to 0.59%, EMA over two and three frames, forward–backward confidence as the weight, aligned-memory propagation, and warped raw previous at H=1; a tuned two-parameter rule on the raw-versus-memory residual, swept over seventeen (α, τ) points, reaches 0.87% at an interior setting. Per backbone, against whichever fixed or EMA policy is best *for that backbone*, the ratio ranges $3.9\times$ to $9.9\times$: the absolute gain depends on the estimator, the margin over smoothing does not.

TABLE II

MATCHED-POLICY CLOSURE ON D2, SEED 0, ALL FIVE BACKBONES AND ALL SEQUENCES POOLED BEFORE TAKING EACH RATIO; REDUCTIONS IN %, HIGHER IS BETTER, DTCE AT LAG k=4. EVERY ROW SHARES THE FROZEN PREDICTIONS, THE ALIGNMENT, THE SUPPORT CONTRACT, THE H=4 STATE MACHINE AND THE EVALUATION CODE — ONLY THE POLICY CHANGES. THE FIRST SIX ROWS LOAD NO CHECKPOINT AT ALL; UNBOUNDED RECURRENCE IS OUR OWN TRAINED HEAD WITH THE RE-ANCHOR REMOVED, WHICH IS WHY IT IS THE ONE THAT INVERTS. THE TUNED RESIDUAL RULE IS THE ONLY UNLEARNED POLICY THAT VARIES ITS WEIGHT IN SPACE, AND THE ONLY ONE THAT DOES NOT PAY FOR ITS MEAN GAIN IN BAD PIXELS. THE FULL GRID OF FIFTEEN POLICIES AND THE SEVENTEEN-POINT SWEEP BEHIND THAT ROW ARE IN THE SUPPLEMENT. TETHER'S ROW IS SEED 0, WHICH IS THE WEAKEST OF ITS THREE SEEDS ON THIS QUANTITY (3.69 AGAINST 4.13 ± 0.38), SO EVERY MARGIN IN THIS TABLE UNDERSTATES IT.

Policy	EPE	Bad1	$\mathrm{DTCE}_{k=4}$	NewBad1
Fixed w=0.5 (≡ EMA2)	0.59	−2.64	6.26	0.214
EMA, three frames	0.30	−8.88	4.88	0.379
FB-confidence as the weight	−2.17	−32.35	−7.10	0.831
Warped raw previous, H=1	−0.03	−6.39	−4.33	0.373
Tuned residual rule	0.87	0.33	2.88	0.088
Aligned-memory propagation	−3.56	−41.70	−13.55	1.072
Unbounded corrected recurrence	−19.17	−143.48	−28.13	2.858
TETHER, H=4	**3.69**	**12.84**	**4.99**	**0.063**
Raw-vs-memory oracle (ceiling)	12.74	21.87	10.01	0.000

Three rows carry more than the mean. Forward–backward confidence is the informative failure — one of TETHER's own cues, yet used directly as the weight it worsens EPE by 2.17% and Bad1 by 32.35%. The uniform smoothing policies buy mean error with bad pixels, the more so the harder they blend: w=0.3 costs 0.18% more bad pixels for 0.49% of EPE, the best fixed blend costs 2.64% for 0.59%, and EMA3 costs 8.88% for 0.30%.

The tuned residual rule is the row that limits that statement, and we report it as such. It is the strongest unlearned competitor, it recovers roughly a quarter of our gain, and it does *not* pay in bad pixels: Bad1 improves by 0.33%. So the trade is not a property of blending, it is a property of blending *uniformly* — the moment a rule varies in space it stops buying mean error with tails. That is the closest any unlearned policy comes to what the head does, and it is also where the distance shows: at 0.87% against 3.69% of EPE and 0.33% against 12.84% of Bad1, it breaks more pixels doing less (0.088% NewBad1 against 0.063%). What the learned head adds is not the idea of varying in space; it is knowing where.

F. Why $H = 4$, and Why Recurrence Must Be Bounded

Recurrence must be bounded: with no re-anchor the intervention inverts, EPE degrading by 19.17% and Bad1 by 143.48% (Table II). Within the bounded range EPE rejects H=1, by 1.24 points, and does not separate the other three, whose spread is 0.21 (3.69, 3.73, 3.52%), while newly broken pixels rise monotonically from 0.052% to 0.128%. We froze H=4 before D7 opened on that risk–gain trade, and it is the last horizon at which change error also improves at lags $k \in \{2, 4, 8\}$ (4.99% at H=4 against -3.64% at H=6 for k=4). Two caveats belong with that. At k=1 DTCE improves at every horizon we tried, so the sign change belongs to the longer lags and not to the metric. And on D2 the improvement is carried by one backbone and is *negative* for three: $+14.0\%$ for Stereo Anywhere and $+0.8\%$ for CREStereo against -0.3, -2.0 and -2.3%. A criterion whose sign one of five backbones decides cannot carry an operating point, so the horizon rests on breakage; on both confirmation splits DTCE is positive for every backbone (Table I).

G. Refinement as an Intervention

Across both splits and all five backbones TETHER recovers 3.8–$11.1\times$ as many Bad1 pixels as it introduces, B^+/H^+ between 1.39 and 3.25, identity preservation above 99.85%. Per frame, 56% of the $15{,}535$ held-out frames improve against 14% worsened (45% against 18% on development), the worsened band hugging the diagonal while the improved mass extends far from it: the count-versus-magnitude asymmetry behind B^+/H^+.

H. What a Map Fuses, Not What a Frame Scores

Everything above scores a frame. What a robot consumes is a map, so we measure that too. SCARED-C's corrected poses let geometry be accumulated in one world frame and the disagreement between views of a point measured — the quantity a reconstruction actually pays for.

On held-out D7 refinement reduces that disagreement by 11.7% of the excess over the ground-truth floor, and the sign is not carried by one estimator: across the full grid of five backbones and four sequences, *all* 20 *cells improve*, by $+7.2$ to $+17.8\%$. The gain tracks how much geometry there is to recover. Raw excess on D7 is 0.518–1.113 mm, so the aligned memory has something a map can fuse; where the raw prediction already sits near the reference floor there is correspondingly less to win, and Sec. VII reports the split where that leaves the intervention a net cost.

I. Head-to-Head Against a Published Stabiliser

BiDAStabilizer [3] is the closest published plug-in stabiliser we could run. We compare on DRENDS because neither method is in domain there; the shared input is its own RAFT-Stereo *robust* prediction, so TETHER is driven on that exact array over all five recordings (7476 frames) under one support neither influences. Two asymmetries run in opposite directions: the baseline is bidirectional and consumes future frames, which our setting forbids, but we are trained on surgical endoscopy while it runs weights fitted to natural-image stereo. We therefore read the margin as an upper bound on our advantage, and an in-domain baseline remains the comparison we cannot run.

On 117.4M pixels of common support, raw EPE 1.199 becomes 1.175 under the baseline and $\mathbf{1.131}$ under TETHER; Bad1 44.92% becomes 44.81 and $\mathbf{43.39}$, and RMSE 1.876 becomes 1.776 and $\mathbf{1.710}$. The regime is hard and those Bad1 rates say so — ground-truth disparity averages 9.89 px, so a raw EPE of 1.199 is a third of the scene's spread — so this compares two refiners of a weak prediction, not a claim that either recovers a good one. TETHER is the only causal entry and carries $62\times$ fewer trained parameters (155k against 9.64M), though at system level we also run frozen SEA-RAFT (8.88M, 26.2 of our 28.9 ms) and make no efficiency claim. The exception is worth naming: on Vid13 TETHER is 0.022 px behind the baseline and 0.013 px *worse than the raw prediction it refines*, the one recording of five where the intervention is a net harm and the bidirectional method is not. In domain the ordering is the same and much wider — D2 EPE $0.3029 \rightarrow 0.2889$ and Bad1 $2.09 \rightarrow 1.80\%$ against 0.3021 and 1.99% — but that margin is measured where we are in domain and it is not.

The temporal measure reverses the ordering, and we report it for that reason. Change error falls 64.9% for the bidirectional baseline against 42.5% for us at lag 1 and 37.2% against 13.8% at lag 4, in each of the five recordings and at every lag; on D2 the split is the same, 52% against 15%. A method that sees the future is markedly more temporally stable than one that cannot, which is what should happen, and it is simultaneously the less accurate of the two on every geometric metric here. That is the paper's own warning turned on its own comparison: a reader given only a consistency score would rank these two the wrong way round.

TC-Stereo [6], the integrated-architecture reference, we run as its authors specify — released SceneFlow weights, its own iteration count, its rigid 6-DoF pose warp on SCARED-C's corrected poses, with the pose convention validated against two wrong alternatives to 0.02 px. On the same D2 support its frame path alone scores EPE 0.524 and Bad1 9.05% against $0.303/2.09\%$ raw, so it is already outside its domain before any temporal component runs; enabling the temporal path makes both worse (0.622, 10.37%) and raises change error from 0.073 to $0.406\,\text{px}$ at lag 1, four to six times *less* consistent at every lag. We read that as a co-designed architecture meeting a scene its rigid warp does not describe, not as a weak method: a TC-Stereo trained on deforming tissue is an experiment nobody has run, ours included. Full protocol, self-check and per-sequence table in the supplement.

J. Ablating Our Own Design Decisions

Four design choices were ablated under a pre-registration naming held-out Bad1 as the endpoint and the three-seed spread (0.37 points) as the threshold below which a difference is not read; A4 remains a single run. A separate later registration governs *promotion* of a variant to the shipped model and decides that on development only, which is how the 38-channel head was promoted; the two rules are distinct and we apply each where it belongs.

TABLE III

HELD-OUT RELATIVE REDUCTION (%), NOT ERROR: HIGHER IS BETTER. THE FOUR DEVIATIONS WERE PRE-REGISTERED AGAINST THE 142-CHANNEL CONFIGURATION; DROPPING ITS LEARNED EVIDENCE IS WHAT TETHER NOW IS, SO THE SHIPPED MODEL IS THE TOP ROW AND THE CONFIGURATION IT WAS ABLATED FROM IS THE ROW BELOW, WITH ITS DEVIATIONS INDENTED. ROWS MARKED "3 SEEDS" ARE MEANS OVER THREE INDEPENDENTLY SEEDED TRAININGS AND CARRY THE STANDARD DEVIATION ON BOTH COLUMNS; ROWS MARKED † ARE SINGLE RUNS. A DIFFERENCE SMALLER THAN THE 0.37-POINT THREE-SEED SPREAD IS NOT INTERPRETED EITHER WAY.

Configuration	Evidence	D7 EPE red.	D7 Bad1 red.
TETHER (3 seeds)	38 ch	$\mathbf{5.61 \pm 0.42}$	$\mathbf{5.84 \pm 0.22}$
+ learned evidence (3 seeds)	142 ch	5.06 ± 0.44	5.15 ± 0.34
no appearance†	78 ch	4.93	4.88
full-res context (3 seeds)	142 ch	5.24 ± 0.52	5.68 ± 0.16
unconstrained output†	142 ch	0.05	6.02

The result (Table III) is not the one we expected, and it is why the shipped model is the smaller one. We began where the video-stereo literature does, with a frozen ResNet-18 [10] supplying appearance correlations and cost curves, 104 channels on top of the 38 geometric ones — the configuration the four deviations were pre-registered against. Three of them sit above it on the held-out endpoint and the fourth, dropping the appearance channels, is 0.27 points below — inside the 0.37-point threshold, so unreadable rather than worse. None of the four decisions is load-bearing. Removing the learned evidence entirely was then given the same three-seed treatment, under the promotion rule that decides on D2 with D7 reported afterwards, and it stays ahead on both endpoints with seed deviation eight times tighter (0.32 against 2.53 points of D2 Bad1). Out of domain the ordering holds: on the three backbones carrying three seeds of each, DRENDS EPE falls by $5.20 \pm 0.13\%$ against $4.83 \pm 0.15\%$, ranges disjoint at a permutation $p = 0.10$. The learned evidence is not what makes the module work; it is what makes its training unstable, at 157,504 frozen parameters, three forward passes per frame and 13.8% of the runtime. Its apparent advantage on the development closure (4.05% against 3.69%) is a seed-0 artefact, and a sharp one: seed 0 is the *best* of three for that configuration and the *worst* of three for ours, so the pairing everyone would report is the worst available to us. Over three seeds the ordering reverses, $4.13 \pm 0.38\%$ for TETHER against $3.55 \pm 0.45\%$.

Two rows need the registered rule read out loud, because it does not favour us. Running the context branch at full resolution reduces held-out Bad1 by $5.68 \pm 0.16\%$ against $5.15 \pm 0.34\%$ over three seeds each: it *wins* the registered endpoint by 0.54 points against a 0.37-point threshold, an outcome the registration did not enumerate. We decline it on cost rather than accuracy — $16\times$ the context-branch compute — which the registration required us to report if this happened. Ablating convexity (A4) posts the largest held-out Bad1 reduction in the table, 6.02%, on one seed; we reject it on EPE, a metric the registration names as reported but not as the endpoint, and that reading was chosen after seeing the result. Its registered consequence we do honour: the falsification clause required that if A4 matched or beat the canonical model held out, the claim attributing cross-backbone transfer to the convex action space be withdrawn rather than softened. It is withdrawn, and appears nowhere in this paper.

K. The Support Contract Is Load-Bearing

Confining the output to (1), not merely computing it, is a single line, measured on the 142-channel configuration, which shares the warp and the mask. On DRENDS the difference is categorical: unmasked, depth RMSE *regresses* in every cell evaluated, by $+110.4\%$; confined, it improves in every cell, by -3.2% (the two arms cover 14 and 15 cells, so the comparison is directional and unpaired). Zero-padded out-of-bounds memory enters the blend as $\widetilde{m}_t = 0$ and bounded recurrence carries the collapse to the next re-anchor; the offenders are rare and bursty (300 of 38.8M pixels on one recording, aligned memory exactly 0.0 for every one) and, since invalid predictions on valid support are penalised rather than discarded, they are the entire source of a depth tail we had attributed to the external domain. On SCARED-C the same ablation is nearly inert: the contract costs a little in-domain gain, buys well-posed behaviour out of it, and is almost invisible to anyone validating in domain only.

VI. ANALYSIS

Alignment is necessary but not sufficient: the aligned memory carries a 12.74% oracle ceiling that indiscriminate use does not collect (Table II). The mechanism is alignment plus learned graded intervention plus bounded recurrence, not smoothing; the head never solves stereo matching, it chooses between two candidates.

a) What each branch of $w_t = r_t f_t$ does.: Suppressing either factor at inference, the trained head unchanged, separates their jobs. Fusion alone keeps most of the gain, 3.49% against 3.69%, but breaks a third more pixels (0.084% NewBad1 against 0.063%); reset alone keeps 2.23% and breaks 0.211%. The fusion weight sets how much is gained,

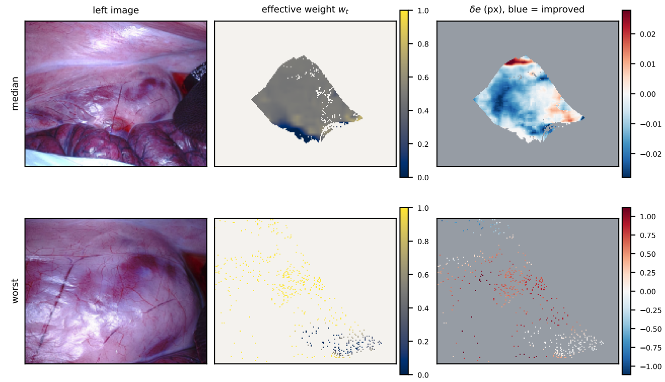

Fig. 2. Where the module intervenes, and whether it helped there. Grey and off-white are off support, where (2) makes the module the exact identity: w_t is undefined and δe is exactly zero, so those pixels are not drawn at the centre of the diverging scale. **Top:** the median frame of a D2 sequence. The module acts on 19.2% of pixels at a mean weight of 0.46 and improves 82.8% of what it touches. **Bottom:** the worst frame of the same sequence, included by rank rather than by choice. It acts on 2.2% of pixels — nine times fewer — at a mean weight of 0.79, and improves 12.9% of them. Note the δe scales differ by $40\times$ between rows (± 0.03 against ± 1.11 px); the failure is few pixels, high weight, wrong, which is the AUROC result of this section drawn rather than tabulated.

the reset gate how little is broken, and a paper reporting only the product and only mean error would have seen neither.

Ablating convexity (A4) collapses mean EPE gain to 0.05% while improving RMSE and worst case: the failure is broad degradation of pixels already close, not occasional large error, so the constraint buys mean accuracy rather than safety from large corrections.

Two limits are worth stating. Against the raw-versus-memory oracle of Table II — measured on TETHER's own aligned memory, since under $H{=}4$ that memory is the previous fused output on three frames of four and the ceiling therefore belongs to whichever head produced it — TETHER recovers 29.0% of the available gain on D2 pooled and 25.2% per cell, so the open problem is deciding *where* to intervene. And the effective weight w_t, recovered from the convex form, is an intervention magnitude rather than an uncertainty — visibly so in Fig. 2, where the frame it damages most is the one it acts on least and most confidently: it ranks harmful intervention at AUROC 0.393–0.408 on D2 and 0.488–0.507 on held-out D7, below chance on the split it was tuned on and at chance on the split it was not. The 142-channel configuration is the same shape (0.366–0.381 and 0.496–0.506), so this belongs to the formulation and not to the evidence space. It does not contradict the intervention ratios, which factorise into a count and a magnitude term needing no per-pixel ranking: a policy can be good on average and uninformative per pixel, which is what makes the oracle gap a ranking problem rather than a capacity one.

VII. Discussion and Limitations

a) What the splits support.: The closure is development work on D2: it supports mechanism analysis and the pre-frozen $H = 4$ operating point, not confirmation. One external domain does not establish universal robustness, DRENDS supplies a temporally filtered ToF reference rather than independent stereo ground truth, and per-backbone D7 margins come from one seed — the three-seed study bounds only the pooled effect.

b) What the metrics cannot say.: No-reference temporal consistency is gameable, which we show on D4D [21], used only as the substrate for that control since its geometric scale contract is unresolved: the module reduces the no-reference score by 53.9%, copy-previous by 97.3%, and warped-previous scores exactly zero by construction. Pull-warp support detects out-of-frame sampling but not internal disocclusion.

c) When there is nothing to recover.: The mapping benefit of Sec. V-H is conditional, and the condition is not subtle. On development D2 the same measurement runs the other way: refinement *increases* world-frame disagreement by 15.5%, and again the sign is uniform — all 15 cells negative, -32.1 to -5.4%. The two splits differ by a factor of two in what there is to remove, raw excess 0.171–0.290 mm on D2 against 0.518–1.113 on D7, which is why we read the benefit as scaling with the size of the raw error rather than as general. We cannot show that this is the mechanism: the two ranges never overlap, so no backbone makes D2 as hard as D7, and within D2 the ordering does not hold either — one estimator loses 32.1% at 0.269 mm of excess and 16.5% at 0.290. What the grid does establish is that the reversal is a property of the split and not of one estimator. A module that refines a prediction already close to its reference can cost a map more than it returns.

d) Cost, and where it is.: The head is compact but the deployed system also runs frozen stereo and two SEA-RAFT passes: 28.90 ms overhead at 95.2 MB peak VRAM on an H100. The frozen estimator dominates, so the end-to-end rate is set by the backbone, 11.6 FPS on Fast-FoundationStereo (overhead 33.5%) to 1.4 on StereoAnywhere (4.0%); *no real-time claim is made*.

Stage timing locates it: the two SEA-RAFT passes are 26.2 of the 28.9 ms, against 0.67 for the evidence tensor and 1.00 for the head. The evidence tensor is that cheap because TETHER builds no learned features — the 142-channel ablation spends 5.13 ms there, 13.8% of its total. The reverse pass exists only to compute C_t^{FB}, so dropping it would free 45% of the module; substituting a constant at inference moves the D2 closure from 3.69% to 3.74–3.79%, well inside the seed spread, so this head appears not to use the cue. We report it rather than ship it: it is development-only, on a head trained with the cue present, and a head trained without it is an experiment we have not run. Details in the supplement. We evaluate perception geometry and what a map accumulates from it, not downstream control, and demonstrate neither clinical safety nor patient benefit.

VIII. Conclusion

TETHER attaches one causal temporal module, trained once, to frozen and heterogeneous stereo estimators without touching their internals or using future frames. Its gain is not temporal averaging: matched fixed and EMA blends under an identical state machine recover a sixth of it and buy that with bad pixels, a trade the learned gate does not make. One frozen $H = 4$ checkpoint improves all five estimators on held-out D7 and all 25 recording–backbone cells zero-shot on DRENDS.

Five results bound what may be claimed, and they are the reason the evaluation protocol is part of the contribution. No-reference consistency is won outright by replaying warped history. The fusion weight inverts its relation to harm between splits, so it is an intervention magnitude and not an uncertainty. Selection optimism on development is a property of the evidence space rather than of the protocol. A module that declares a support must apply it to what it emits. And what a map fuses improves held out on every backbone we tried while degrading on development, so the benefit scales with how much geometry there is to recover rather than being general. What remains open is *where* to intervene.

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